Nonnegative and positive factorizations

The maximum cp-rank in order six

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Historical ratings for the original open question: challenging because sharp cp-rank control was missing for the remaining order-six boundary configurations; community importance is the size of exact completely positive certificates.

Area: size of nonnegative symmetric factorizations

Resolution — affirmative, 13 September 2026 (UTC)

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Theorem 1.1, proved in Sections 2–8, proves the full retained target: every real completely positive matrix of order six has a nonnegative factor with at most nine columns, including singular matrices and zero entries. Proposition 8.1 gives a positive-definite matrix attaining nine, so $p_6 = 9$. Editable proof.

The complete argument passed a separate independent Codex AI-agent informal audit. Exact auxiliary checks were rerun successfully. This meets the repository's Solved policy; it is not external human peer review or formal verification. AI assistance in preparation and review is disclosed. No Lean verification was performed. The original target, ID, canonical path and prior-source credit are retained below as historical context.

Context and notation

A symmetric matrix A is **completely positive** if $A = BB^{\mathsf{T}}$ for an entrywise nonnegative real matrix B with finitely many columns. Let $\mathcal{CP}_n$ be the cone of these matrices of order n. Its **cp-rank**, $\operatorname{cpr}(A)$, is the minimum number of columns in such a factor, with $\operatorname{cpr}(0) = 0$. Boundaries and interiors use the usual Euclidean topology on the vector space of real symmetric matrices.

Problem statement

Is every real completely positive matrix of order six a sum of at most nine nonnegative rank-one matrices? Equivalently, is

$$\max_{A \in \mathcal{CP}_6} \operatorname{cpr}(A) = 9,$$

or, equivalently, does each $A \in \mathcal{CP}_6$ admit $A = BB^{\mathsf{T}}$ with $B \in \mathbb{R}^{6\times 9}_{\geq 0}$, allowing zero columns?

The lower bound nine is established. Order six is the unresolved case of the original Drew–Johnson–Loewy bound: order five satisfies it, whereas counterexamples exist in higher orders. It is included as the specific remaining case identified in the literature, rather than as one item in a dimension-by-dimension list.

A known reduction places a matrix attaining the order-six maximum on the boundary of $\mathcal{CP}_6$, with full ordinary rank and at least one zero entry. This reduction does not establish the nine-column bound.

References

Abraham Berman, Mirjam Dür, and Naomi Shaked-Monderer, *Open problems in the theory of completely positive and copositive matrices*, Electronic Journal of Linear Algebra **29** (2015), 46–58, §4.2, p. 53. Naomi Shaked-Monderer, *On the DJL conjecture for order 6*, corrected arXiv v3, 2017-06-01; journal version in Operators and Matrices **11** (2017), 71–88, Theorem 1.1 and Corollary 3.2.

Status check — 2026-09-10

Rechecked Shaked-Monderer's corrected v3, Theorem 1.1 and Corollary 3.2, and searched for later order-six DJL resolutions. The nine-term bound is proved on the boundary classes orthogonal to exceptional extremal copositive matrices, including positive nonsingular boundary matrices. Remaining maximizers can have a zero entry; this reduction is not a proof for all order-six matrices. Behling et al. (2026), §5.6, reproduces counterexamples in orders seven and twelve, not six. No full resolution was located, and the exact open-status source remains historical.